Referee report on `mjl_part4_draft1.tex`

"Toward a metaplectic Jacquet–Langlands transfer, IV"

Claude (Anthropic), for David Joyner

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0. Scope and what I read

Read in full. Part IV, Draft 1 (669 lines), and its log; Part III, Draft 6 (1359 lines); the Part III log (Entries 0–6); my Draft 5 report on Part III; the Part II log; and the revision log for Part I Draft 6 and [JC] revised12.

Read in part.

- *Part I, Draft 6:* lines 1–949, i.e. §§1–8 (everything except the glossary, literature section and Appendices A, B).
- *Part II, Draft 6:* the introduction and dependency diagram (lines 1–140), Definition 2.1 and Remark 2.2 (measures), §7 ("What is left for Part III"), and the acknowledgments. I did not check Part II's proofs here.
- *[FK] (scan) and Zou (arXiv v3):* I extracted the text and read only the passages listed in §3 below.

Not read. [JC] (`metaplectic_kazhdan_revised12.tex`): I compiled it but did not read it, since Part IV does not cite it.

Compilation. All of Parts I–IV compile. In my build the page counts are 48, 32, 23 and 11. The theorem numbers I quote below come from the compiled `.aux` files. Part IV's numbering is as follows (equations are numbered consecutively in all four papers):

- Proposition 2.1 (lattice criterion);
- Remark 2.2;
- Proposition 2.3 (finite-level test);
- Remarks 2.4, 2.5;
- Lemma 3.1 (radical);
- Theorem 3.2 (split diagonal good elements);
- Remarks 3.3, 3.4;
- Definition 3.5 (nonsplit-torus problem);

- Proposition 4.1 (Hasse principle);
- Remark 4.2;
- Definition 5.1 (geometric comparison);
- Remark 5.2;
- Proposition 6.1 (spectral).

Summary verdict

I found no false mathematical identity in Part IV. Every displayed formula I checked is correct. The draft's main problem is that it does not take account of what Part I and [FK] already contain.

(1) *The split-place theorem and the "open" nonsplit problem are already in [FK].* Theorem 3.2 is [FK, §3, Proposition] specialized to the diagonal torus. That Proposition treats *every* regular torus $\bigoplus_i F_i^\times$, at every finite place, tame or wild. So Definition 3.5 is not an open problem. Part I already uses the Proposition: in the proof of Theorem 5.6, and through its commutator formula in Proposition 3.1.

(2) *The global-comparison sections ignore Parts I and II.*

- Part I already contains a proved trace-formula comparison: Theorem 6.6, with rational-orbit accounting (Theorem 5.6, Corollary 5.7), measures (§5.3), and identity terms (Proposition 6.1). Part II extends it (Theorem 6.2).
- Part IV's introduction, status table, §4 and §5 describe all of this as open.
- Definition 5.1 does not say *which* two groups are being compared. Part I compares $\overline{G}_D$ with $\overline{G}$. The target identity (31) of Part III compares $\overline{G}_D$ with $D^\times$.

(3) *The draft misses a new global result that it is close to.* With (1), Proposition 4.1 *already* identifies the globally good regular elliptic classes. Remark 4.2 says this is not yet possible. I propose a short replacement corollary in §1.4.

(4) *Section 2 adds little to Part III.* Propositions 2.1, 2.3 and 6.1 restate Part III, Corollaries 5.1 and 5.2. The "inverse" conditions in them are redundant, because Part III, Theorem 4.3, proves that L_D is a unitary *isomorphism*.

Once these points are dealt with, the paper's genuinely new content is small: the corollary in §1.4, the finite-level wild test, and a precise formulation of the $\overline{G}_D$–$D^\times$ comparison. I recommend restructuring along those lines (§5) before any further refereeing.

Result	Mark	Main issue
Prop. 2.1 (lattice criterion)	correct; minor repair	Its hypotheses are Part III theorems; the inverse clause is redundant; it is Part III, Cor. 5.2, plus a trivial converse
Rem. 2.2	misleading	For a unitary L_D, the inverse condition is automatic
Prop. 2.3 (finite level)	correct	This is Part III, Cor. 5.1
Rem. 2.5 (Adams)	correct; minor repair	The relevant condition is n prime to $\lvert Q/C\rvert$, not to $\lvert Q\rvert$
Eq. (7), Lemma 3.1	correct; minor repair	c should be c_v; "alternating" is misnamed; $(r,n)=1$ is automatic
Thm. 3.2	correct	It is [FK, §3, Prop.] for the diagonal torus; label **[E]**
Def. 3.5 **[O]**	status false	Already solved by [FK, §3, Prop.] (see §1.3)
Prop. 4.1	correct	This is Part III, Prop. 11.3; cite the number
Rem. 4.2	incorrect claim	The identification it calls missing is in [FK, §3]; replace it by the corollary in §1.4
§5.1 (three quantities)	minor repair	b_D and c_D do not depend on the choice of measures (§1.5)
Ledger, Def. 5.1	serious gap (formulation)	The groups compared are unstated; several rows are settled in Part I
Prop. 6.1	correct	This is Part III, Cor. 5.2 again; the inverse clause is redundant

1. Correctness

1.1 Section 2: the character lattice

Proposition 2.1: correct, minor repair.

- *The hypotheses are not hypotheses.* "Unitarity, smoothness, and finite character expansions" are proved in Part III: Theorem 4.3, Lemma 4.2 and Corollary 5.1. Assuming them makes the result look conditional. (I raised the same point about "Hypothesis (G)" in my Draft 5 report on Part III.)
- *The inverse clause is automatic.*
  - Since L_D is a unitary isomorphism and both character families are orthonormal bases, the coefficient of Θ_ρ in $L_D^{-1}\Theta_\tau$ is $\overline{a_{\rho\tau}}$.
  - If every $a_{\rho\tau}\in\mathbb{Z}$, each row is a signed basis vector. Distinct rows are orthogonal, so $\rho\mapsto\tau(\rho)$ is injective.
  - Every column has norm 1, so every τ is hit.

  Hence (iii) can be stated as "every $a_{\rho\tau}$ is an integer". The sentences about the adjoint matrix should then be deleted.
- *Remark 2.2 should be rewritten.* As it stands it suggests a real issue that unitarity already removes.
- *Status.* The implication (iii)⇒(i) is Part III, Corollary 5.2. The converse is trivial. I suggest the label **[E]** (Part III, Cor. 5.2) with a one-line proof of the converse, not a new label **[R]**.

The sign argument is correct: $(L_D\Theta_\rho)(1)=c_D\dim\rho>0$ by Part III, Corollary 5.2.

Proposition 2.3 and Remark 2.4: correct. This answers log question (Q3) affirmatively, and the argument is already the proof of Part III, Corollary 5.1. A smooth $\tilde{\theta}$-equivariant invariant function is right-invariant under an open normal subgroup N, because $\widetilde{H}/\widetilde{Z}$ is compact. Only the finitely many τ with $\tau^N \neq 0$ and central character $\tilde{\theta}$ can occur, since $N\widetilde{Z}$ has finite index. I suggest citing Part III, Corollary 5.1, and keeping only the computational remark.

Remark 2.5 (Adams operations): correct, minor repair. The sentence "when n is prime to $|Q|$ ... this is the ingredient used in the tame case of Part III" misdescribes Part III, Lemma 7.5. That lemma allows a central subgroup C of order divisible by n, with the central character fixed, and needs only $\gcd(n, |Q/C|) = 1$. The obstruction at a wild place is therefore $p \mid |Q/C|$. This does occur:

- every finite quotient of H/Z maps onto a quotient of $E^\times/F^\times E^{\times n}$ for a maximal subfield E;
- that group has order $|n|_F^{-(r-1)}$ (Part III, Lemma 3.2).

The C_p example is right if read with $C = 1$. It should say so.

1.2 Section 3: the commutator matrix

Equation (7) and Lemma 3.1: correct. I recomputed $\sigma_c(t,u)/\sigma_c(u,t)$ directly from (5). Antisymmetry $(a,b)(b,a) = 1$ gives

$$\prod_{i,j}(t_i, u_j)^{2c+[i\neq j]}, \qquad \text{so} \qquad A = -I + (1+2c)J.$$

This agrees with Part I, Proposition 3.1. That proposition, following [FK, §3] and [KP1, Prop. 0.1.5], gives $[x,y] = (\det x, \det y)^{1+2c_v}\prod_i (x_i, y_i)^{-1}$ on any maximal torus. The kernel computation in Lemma 3.1 is correct. Four repairs:

(a) *Notation.* Write c_v for the parameter. In Part I (Assumption 1.1) the letter c denotes the integer with $n \mid r - 1 + 2c$, and $c_v = r'c$.

- Part IV's c in (5)–(6) is Part I's c_v. The example $c \equiv 2 \pmod 3$ matches Part I, Example 3.7, which has $c = 1$ and $c_v = 2$.
- Since $2r$ is invertible modulo n, the congruence determines c_v modulo n. Part I, §5.1, notes that c_v does not depend on v. So the ledger row "dependence on c_v" is vacuous.

(b) *Terminology.* "Since n is odd, the Hilbert symbol is alternating: $(x,y)_n = (y,x)_n^{-1}$." The displayed identity is antisymmetry, which holds for every n. "Alternating", meaning $(x,x) = 1$, is what needs n odd, and it is not used.

(c) *A redundant hypothesis.* If $n \mid r - 1 + 2rc$ then $\gcd(r,n)$ divides 1, so $(r,n) = 1$ is automatic.

(d) *Matching Part I's cocycle.* Log question (Q1) asks whether (5) agrees with Part I. It does, and literally, not only up to a coboundary, in the σ-model after relabeling. Part I, Proposition 2.3(b) and Step 1 of its proof, show that $\overline{G}_v$ is the extension with cocycle $\sigma^r(\det,\det)^c$. On the diagonal torus this is $\prod_{i<j}(a_i, a_j')^r(\det,\det)^c$. After the relabeling $\zeta \mapsto \zeta^{r'}$ it is exactly (5) with $c = c_v$. Before relabeling, every commutator is raised to the power r, and the radical is unchanged.

Theorem 3.2: correct, but known. The statement is [FK, §3, Proposition]: "$[x, g] = 1$ for all g in T if and only if x lies in Z_*T^n." I checked this on p. 59 of the attached scan. There Z_* is the inverse image of the center of the cover, and $Z_* = Z$ by Part I, Lemma 2.2. The theorem should be labeled **[E]** and attributed. The abstract's "we compute the commutator radical" should become "we recall".

1.3 Definition 3.5 is not open

[FK, §3] states the commutator formula for the torus $T = \bigoplus_i F_i^\times$ of an arbitrary regular element: $[x, g] = (\det x, \det g)^{1+2c} \prod_i (x_i, g_i)_{F_i}^{-1}$. It deduces goodness $\iff$ $x \in Z_*T^n$ for every such torus. The proof is three lines, and it is the proof of Part I, Proposition 3.5(a), word for word.

- Write $E = \prod_i E_i$ and $N = N_{E/F}$.
- Norm compatibility gives $(\det x, \det g)_F = \prod_i (Nx, g_i)_{E_i}$.
- Nondegeneracy of each $(\cdot,\cdot)_{E_i}$ (valid at every finite place, since $\mu_n \subset E_i$) gives: x is good $\iff$ $x_i \equiv (Nx)^{1+2c_v} \bmod E_i^{\times n}$ for all i.
- The congruence $n \mid r - 1 + 2rc_v$ turns this into $x \in F^\times \prod_i E_i^{\times n}$, exactly as in Part I.

No new "restriction and corestriction" analysis is needed, and the answer is uniform in E. Places with $p \mid n$ are included. So Definition 3.5 should be deleted. The paragraph after it, warning that rational points need separate treatment, is moot: the formula is stated on F-rational points.

1.4 Section 4: what Proposition 4.1 already gives

Proposition 4.1: correct. This is Part III, Proposition 11.3; I rechecked the algebra $t/a = (a_v^k y_v N(y_v)^{-u})^n$. Log question (Q5): the hypotheses used are

- $[E : K] = r$ (in the norm trick);
- $\gcd(n, r) = 1$ (to choose u, k);
- $\mu_n \subset K$ (Kummer theory over E).

n odd is not used.

Remark 4.2 is wrong in its key claim. It says the proposition "cannot be invoked from the good-element condition unless that condition has first been identified with the local power condition for the actual regular centralizer". By §1.3 that identification holds at every place. I suggest replacing the remark by the following statement, which I believe is new in the series and is the rational-orbit accounting asked for in Part III, §12, obligation (1).

Proposed Corollary ([P]). Assume Part I, Assumption 1.1. Let $G \in \{G_D, \mathrm{GL}(r)\}$, and let $\gamma \in G(K)$ be regular elliptic, with $E = K(\gamma)$ a field of degree r. The following are equivalent:

(a) $\iota(\gamma)$ is good at every place;

(b) $\iota(\gamma)$ is good at every place outside some finite set Σ;

(c) $\gamma \in K^\times E^{\times n}$.

Moreover $s \mapsto s^n$ is a bijection from the regular elements of $E^\times/K^\times$ onto the classes of $E^\times/K^\times$ satisfying (a)–(c), and $W_{s^n} = W_s$ in the notation of Part I, §6.

Proof sketch.

(i) *The local criterion at each place.* At a finite place v, the centralizer of γ is $T_v = (E \otimes_K K_v)^\times$.

- For $v \notin S$, combine Part I, Proposition 2.6 (on the D side), Proposition 2.3(b), Lemma 2.2 and [FK, §3]: goodness at $v \iff \gamma \in K_v^\times T_v^n$.
- For $v \in S$, the same holds by Part I, Proposition 3.5(a).
- At the complex places everything is split and $T_v^n = T_v$.

(ii) *The equivalence.* (c)⇒(a)⇒(b) are clear, and (b)⇒(c) is Proposition 4.1.

(iii) *Injectivity of $s \mapsto s^n$.* If $s^n \in K^\times$, then $[K(s) : K]$ divides both n and r, so it is 1.

(iv) *Regularity.* If as^n is regular then s is not central.

(v) *The groups W.* For $\sigma \in \mathrm{Aut}(E/K)$, $\sigma(s^n) \in K^\times s^n \iff (\sigma(s)/s)^n \in K^\times \iff \sigma(s) \in K^\times s$, by the same injectivity. □

So the geometric sides in Part I, Propositions 6.1 and 6.2, run over $\{[s^n]\}$, and the global "Kummer obstruction" disappears for the comparison of $\overline{G}_D$ with $D^\times$.

§4.1 (Kottwitz): correct. This is consistent with Part III, §10.

1.5 Section 5: normalizations

- *The formulas are right.* The formulas for $b_{D,v}$ and $c_{D,v}$ agree with Part III, Theorem 4.3, Corollary 5.2 and Corollary 9.3. Log question (Q2): yes.
- *But b_D and c_D do not depend on the measures.* Part III identifies $\widetilde{H}/\widetilde{Z} \cong H/Z$ canonically. The covered orbital integral at zt^* is an integral over the same quotient H/T as the linear one at t. So if the same Haar measure dg is used on H/Z and $\widetilde{H}/\widetilde{Z}$, and the same dt on T/Z:
  - both orbital integrals in Part III, (20), are multiplied by the same factor, the ratio of $\mathrm{vol}_{dg}(H/Z)$ to $\mathrm{vol}_{dt}(T/Z)$;
  - both L^2 norms in Theorem 4.3 are multiplied by the same constant.

  Thus (20), (21) and unitarity hold with b_D for *every* common choice of measures. The value c_D at the identity is pointwise, so it is measure-free. The "third contribution" in §5.1 is therefore zero for a comparison of $\overline{G}_D$ with $D^\times$. Measures matter only in the comparison of $\overline{G}_D$ with $\overline{G}$, where Part II, Definition 2.1, fixes $\mathrm{vol}(D_v^\times/F^\times) = d(\mathrm{St})^{-1}$. Under that choice the probability measure is $d(\mathrm{St}; dg_v)\, dg_{D,v}$.
- *The product-formula remark is correct.* Here is a concrete point for the ledger. At a place $v \notin S$ where $E \otimes K_v$ has k_v factors, the map $s \mapsto s^n$ on T_v/Z_v is *not* injective. Its kernel is $\mu_n^{k_v}/\mu_n$, of order n^{k_v-1}, even when $v \nmid n$.
  - Contrast $v \in S$: there the power map is injective (Part III, Lemma 3.2).
  - This is the reason for the sum over $\{t : t^n = x^n\}$ in Zou's (3.2).

The local transfer at split places, and the bookkeeping of these kernels against the single global s of §1.4, must be stated explicitly.

- *The ledger rows.* Several are already settled.
  - "Global cocycle": Part I, Lemma 5.1 and Proposition 5.5 (reciprocity, rational and compact splittings).
  - "Global torus volume": Part I, §5.3, uses one measure on $\mathbb{A}_E^\times/\mathbb{A}_K^\times$ on both sides.
  - "Local orbit": §1.4 above.
  - "Singular and identity terms": Part I, Proposition 6.1, computes the identity term; regular support at one place kills it; Part II, Theorem 5.13, treats the germ at the center.
- *Definition 5.1 must name the groups.* If the target is Part III, (31), i.e. $\overline{G}_D(\mathbb{A}_K)$ against $D^\times(\mathbb{A}_K)$, then:
  - the $\overline{G}_D$ geometric side is Part I, Proposition 6.1;
  - the $D^\times$ side is the classical one;
  - the orbit bijection is §1.4;
  - at $v \in S$ the local identity is Part III, Theorem 9.1;
  - what remains is (i) the local transfer at $v \notin S$ between $\overline{G}_v$ and $\mathrm{GL}(r, K_v)$, which is the Flicker–Kazhdan matching, with unit elements under $(n, N) = 1$ [FK, Corollary of §12], and (ii) the product of all local constants.

  That is a far more precise open problem than the present Definition 5.1.
- *Remark 5.2.* Add, as Part II, §7(iii) does, that [Li4] assumes an invariant Paley–Wiener theorem at the archimedean places.

1.6 Section 6

Proposition 6.1: correct. Given integral m_τ, unitarity gives $\sum m_\tau^2 = 1$, and positivity at 1 fixes the sign. This is Part III, Corollary 5.2, applied to one row. The hypothesis "the inverse coefficients are integral" is redundant (§1.1). The proposition is labeled **[R]**, but no reduction is proved. It is a restatement.

2. Consistency with Parts I–III

(1) *Description of the earlier parts.* §1 says Part I develops "the specific covering cocycles, their centers, and good elements". In fact Part I proves Theorems A, B and C:

- Theorem 3.9;
- Theorem 4.5;
- the trace-formula comparison, Theorem 6.6;
- the packet identity, Proposition 7.6;
- Theorem C itself, Theorem 7.11, under (H_D).

Part II proves Theorem E (Theorem 5.8) and Theorem C′ (Theorem 6.5, under (H'_D), Definition 6.4). The status-table row "Global trace-formula coefficients: **[O]**" contradicts Part I, Theorem 6.6 **[P]**. It should refer specifically to the $\overline{G}_D$–$D^\times$ comparison.

(2) *Cross-references.* Part IV never cites a numbered result of Parts I–III. Suggested citations:

- L_D: Part III, Def. 4.1 and Thm. 4.3.
- Finite expansions: Part III, Cor. 5.1.
- The identity value and c_D: Part III, Cor. 5.2, Lemma 5.4 and Cor. 9.3.
- The factor b_D in dimensions: Part III, Prop. 8.1.
- The tame case: Part III, Lemma 7.5 and Thm. 7.6.
- The Hasse principle: Part III, Prop. 11.3.
- Scalar centrality: Part I, Lemma 2.2.
- The cocycle: Part I, Prop. 2.3 and eq. (1).
- The torus commutator: Part I, Prop. 3.1.
- Elliptic good elements: Part I, Prop. 3.5.

(3) *Central characters.* §2.1 says "fix θ ... and the corresponding genuine $\tilde{\theta}$". Part III goes the other way: it fixes $\tilde{\theta}$ and puts $\theta(a) = \tilde{\theta}(a^*)$. Since $\theta = \tilde{\theta}^n$ on lifts, θ does not determine $\tilde{\theta}$. Fix $\tilde{\theta}$ first, and write $L_D^{(\tilde{\theta})}$ where it matters.

(4) *Labels.* Part I defines **[O]** as "proof outlined" and **[C]** as "conjectural". Parts III and IV use **[O]** for "open problem" and **[C]** for "conditional". Part IV adds **[R]**. One convention should be adopted for the series. I would drop **[R]**.

(5) *The sign $(-1)^{r-1}$ and the link to Part I.* Part IV works only on the division side and never mentions the GL-side character relation. Your consistency report says that the equality $d_r = n$ makes Zou's constant in (3.2) equal to b_D. Then Part I's $\Theta_{D} = (-1)^{r-1}\Theta \circ \jmath$ (goal (D2); Part II, §7(ii)) is equivalent to integrality of L_D. If so, the "wild local correspondence" of Part IV, §2, *is* Part II's goal (ii) at wild places of S, and the paper should say so. I checked two ingredients of your report against Zou:

- Zou, (2.2), gives the center as $p^{-1}\{\lambda : \lambda^{2rc+r-1} \in F^{\times n}\}$, the same convention as ours;
- Zou, §2.3, defines $d_{r_0} = \gcd(n, 2r_0c + r_0 - 1)$.

So $d_r = n$ under $n \mid r - 1 + 2rc_v$. Note, however, that a later passage of Zou (p. 18 of arXiv v3, §4, just after the proof of Proposition 4.2) writes $d_r = \gcd(n, 2rc - r + 1)$. This appears to be a typo in Zou, but it is worth a footnote.

(6) $(n, N) = 1$. [FK, Introduction and Corollary of §12] assume $(n, N) = 1$, with N the least common multiple of the composite integers $\leq r$. Part III, Remark 12.1, records this. Part IV does not mention it, although any use of Flicker–Kazhdan matching at split places (§1.5) needs it unless Zou or later work removes it.

(7) *Measures and Δ.* Part I's $\Delta = |D(\gamma)|^{1/2}|\det\gamma|^{-(r-1)/2}$ (Definition 4.1) equals Part III's product (3), since $\prod_{i<j}\lambda_i\lambda_j = \det^{r-1}$. This is consistent. For measures, see §1.5.

(8) *Titles and authors.* The log says the title is "aligned with the rest of the series". It is aligned with Part III only. Parts I and II are titled "A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, I/II". The author lines also differ:

- Parts I, II: "David Joyner and Claude";
- Part III: Joyner, with assistance from Claude and ChatGPT;
- Part IV: Joyner, with assistance from ChatGPT only.

This is your decision. The four papers should be made consistent.

3. Citations

Checked against the attached sources.

- *[FK86].* Publ. Math. IHÉS 64 (1986), pp. 53–110.
  - The page range is confirmed by the scan.
  - §3, Proposition, with the torus commutator formula credited to [KP, Prop. 0.1.5].
  - §9, Lemma and Proposition (good elements of the form x^n).
  - The Introduction and the Corollary in §12 for $(n, N) = 1$.
- *Zou, arXiv:2209.05852v3 (2024).*
  - (2.2) (center);
  - §2.3 (d_r);
  - (3.2) and Remarks 3.1 and 3.8 (the corrected scalar);
  - Proposition 3.6 (built on [FK, Theorem 26.1]);
  - the standing assumption $\gcd(n, p) = 1$ in §4.

  Part IV does not cite this paper at all. It should, because the relation of L_D to the GL-side correspondence goes through Zou's (3.2) (compare Part III, Remark 4.4).

Not verified by me (sources not attached).

- *[ZouTypes]* (J. Inst. Math. Jussieu 24 (2025)): the bibliographic data, and the claim that it assumes $(p, n) = 1$.
- *[Takeda]* (Pacific J. Math. 290 (2017)): the data, and what "the scalar-center formula" refers to.
- *[Kottwitz84], §§5, 7–9:* the Part III log reports these checked.
- *[Li14] = [Li4]:* the Part II log reports it checked on EuDML. I have not checked the DOI.
- *[DKV].*

[Cohen] and [Adams] are in the bibliography but are never cited.

4. Readability

(R1) *Drafting history must go to the log.* Three passages are about the drafting process, not the mathematics:

- the end of the "status labels" paragraph ("Claude's proposed cross-part review is therefore a particularly valuable prerequisite ... ");
- the acknowledgments' "whom the author plans to consult separately ... ";
- the acknowledgments' "should be checked against the exact cocycle conventions of Part I before publication".

After §1.2(d) above, the last of these is also resolved.

(R2) *Disclaimers.* "No ... is claimed" appears in the abstract, §1, the remarks after Theorem 3.2, §4, §5, the "preferred order of work" paragraph and the acknowledgments. State the status once, in the table, as Parts I and II do.

(R3) *Orienting paragraphs.* These are mostly present. §2.1, §3.1 and §4.1 begin abruptly.

(R4) *Glossary.* As I said for Part III: entries such as Hilbert symbol, central character, induced torus and Tamagawa measure read as padding in a research paper. Keep only a short notation list of terms specific to the paper.

(R5) *Abstract.* "We compute the commutator radical ... and obtain an explicit good-element criterion" should become a citation of [FK, §3].

(R6) *One running example.* The two numerical remarks for $(2, 3, \mathbb{Q}_3(\zeta_3))$ should be merged and should cite Part I, Example 3.7, which uses the same data.

5. Answers to the log's questions, and recommendations

(Q1) Yes: §1.2(d).

(Q2) Yes, and the constants do not depend on the measures: §1.5.

(Q3) Yes: §1.1.

(Q4) Part II's matching relates $\overline{G}_D$ to $\overline{G}$. Part III's pairing relates $D^\times$ to $\overline{G}_D$. They are different transfers, so no compatibility between them is needed. When the two are composed, use Part II's measures on the GL side and any common measure on H/Z (§1.5).

(Q5) Yes: §1.4.

(Q6) Yes: [FK, §3, Proposition] eliminates the nonsplit-torus task entirely (§1.3).

Suggested Draft 2.

1. An introduction that accurately states what Parts I–III prove.
2. The wild problem as Part II, goal (ii), at wild places, via Zou's (3.2) and your consistency report. Proposition 2.1 is reduced to a remark citing Part III, Corollary 5.2.

3. Good elements:

FK, §3 for all tori, labeled [**E**];

- the corollary of §1.4, labeled [**P**].

4. A precise statement of the $\overline{G}_D$–$D^\times$ comparison:

   - the geometric sides, from Part I, Proposition 6.1;
   - the local identities required at each place;
   - the product of local constants, including the kernels $\mu_n^{k_v}/\mu_n$ at split places;
   - $(n, N) = 1$.

   Proving this would be a real theorem. Even a complete statement of the reduction would be useful.

5. The explicit test for $(2, 3, \mathbb{Q}_3(\zeta_3))$, carried out.

   - Here $c_D = 27$. With ρ trivial, $L_D\Theta_\rho$ is $3R(t)\tilde{\theta}(z)$ on zt^* and 0 elsewhere.
   - Computing its coefficients needs a finite quotient $\widetilde{H}/\widetilde{Z}U_k$ and its genuine character table.
   - This is a feasible GAP or Sage computation, and it decides the first case of the wild problem.